A Systematic Viewpoint of Geometry-aware Filtering for Nonlinear Systems

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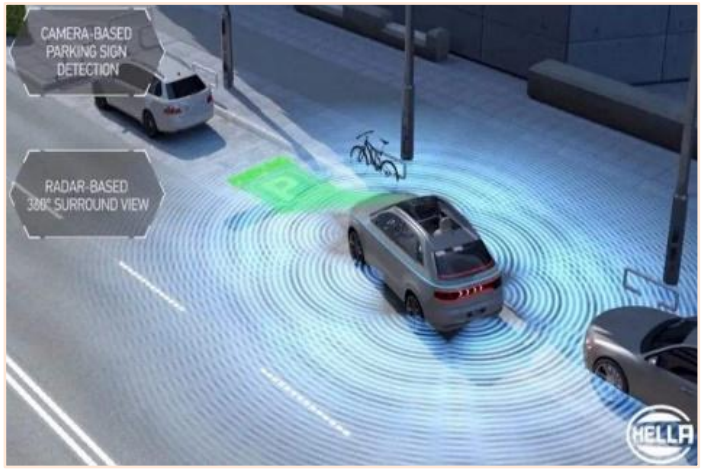
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Motivation

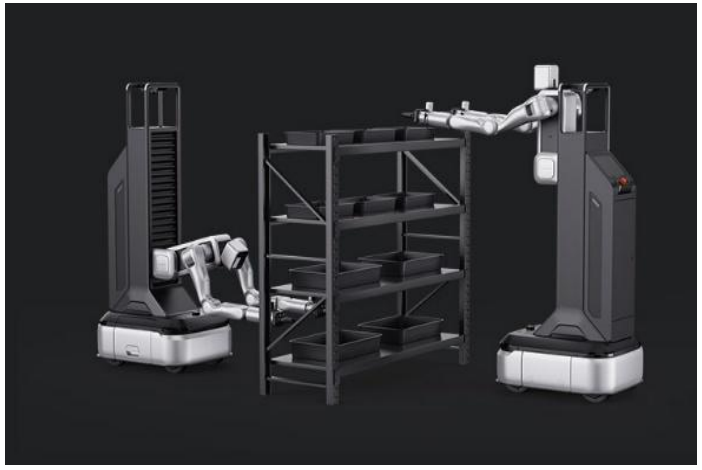
- **State estimation is indispensable in science and engineering**













History of Filtering Algorithms

- **Bayesian filtering builds theoretical foundation of state estimation**

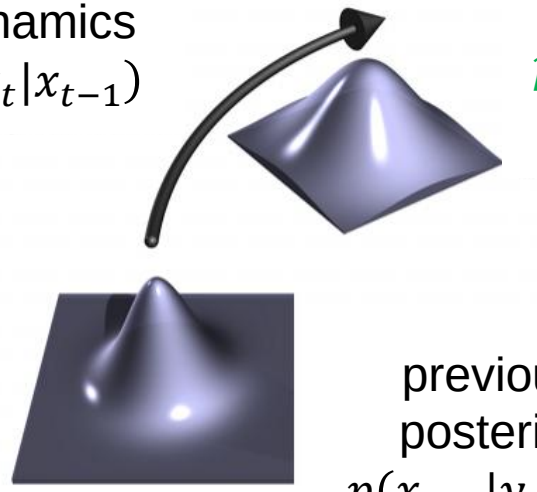
Frequency Domain Linear	Temporal Domain Linear	Discrete-time Nonlinear	Discrete-time Nonlinear	Continuous-time Nonlinear
				
N Wiener	R E Kálmán	S Julier	A Doucet	S ST Yau
Wiener Filter	**Kalman Filter**	**Unscented KF**	**Particle Filter**	**Yau-Yau Filter**
(1949)	**(1960)**	**(1997)**	**(1995)**	**(2008)**

Basis of Bayesian Filtering

- **Two steps of Bayesian filtering: Prediction and Update**

Prediction

dynamics
$p(x_t|x_{t-1})$



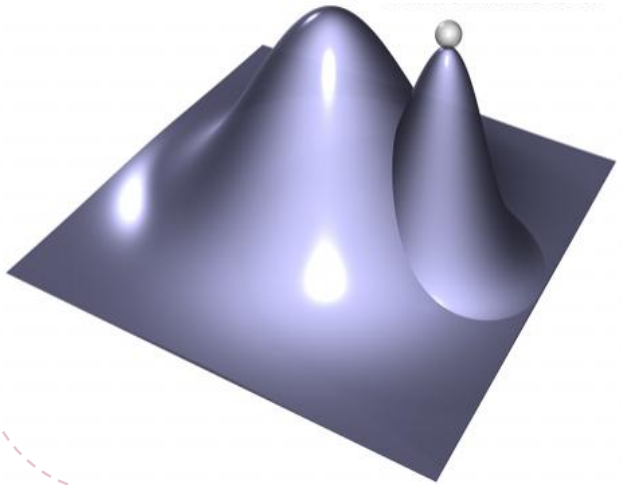
prior
$p(x_t|y_{1:t-1})$

previous
posterior
$p(x_{t-1}|y_{1:t-1})$

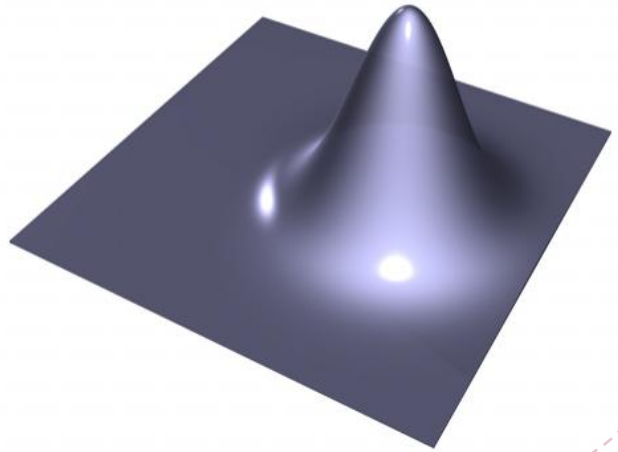
Update

prior
$p(x_t|y_{1:t-1})$

likelihood
$p(y_t|x_t)$



posterior
$p(x_t|y_{1:t})$



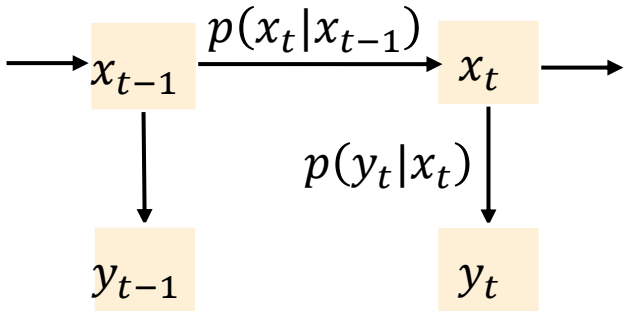
Propagate uncertainty through transition model

Incorporate new observation to refine belief

Basis of Bayesian Filtering

Procedure of Bayesian filtering

Step 1: Build model



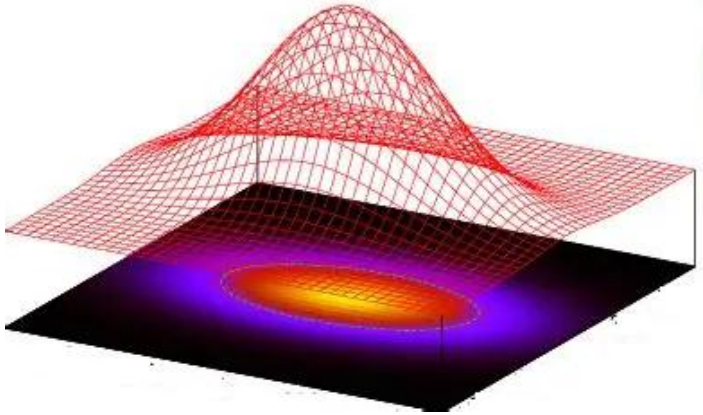
Step 2: Calculate PDF of prior and posterior

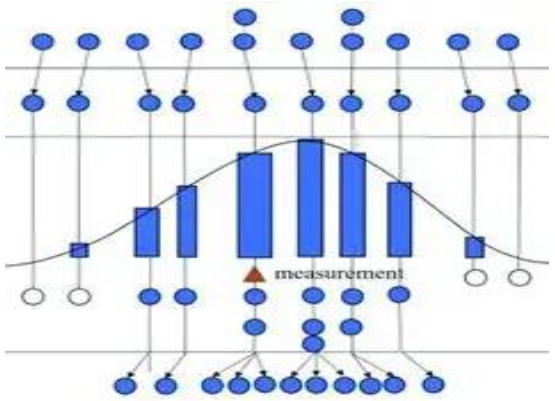
Prediction $p(x_t|y_{1:t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|y_{1:t-1})\,\mathrm{d}x_{t-1}$

Update $p(x_t|y_{1:t}) = \dfrac{p(x_t|y_{1:t-1})p(y_t|x_t)}{\int p(x_t|y_{1:t-1})p(y_t|x_t)\,\mathrm{d}x_t}$

① **Gaussian representation**
Approximate PDF as Gaussian

② **Particle representation**
Approximate PDF as Particles





Step 3: Point estimate

Estimation Criterion

① Minimum MSE

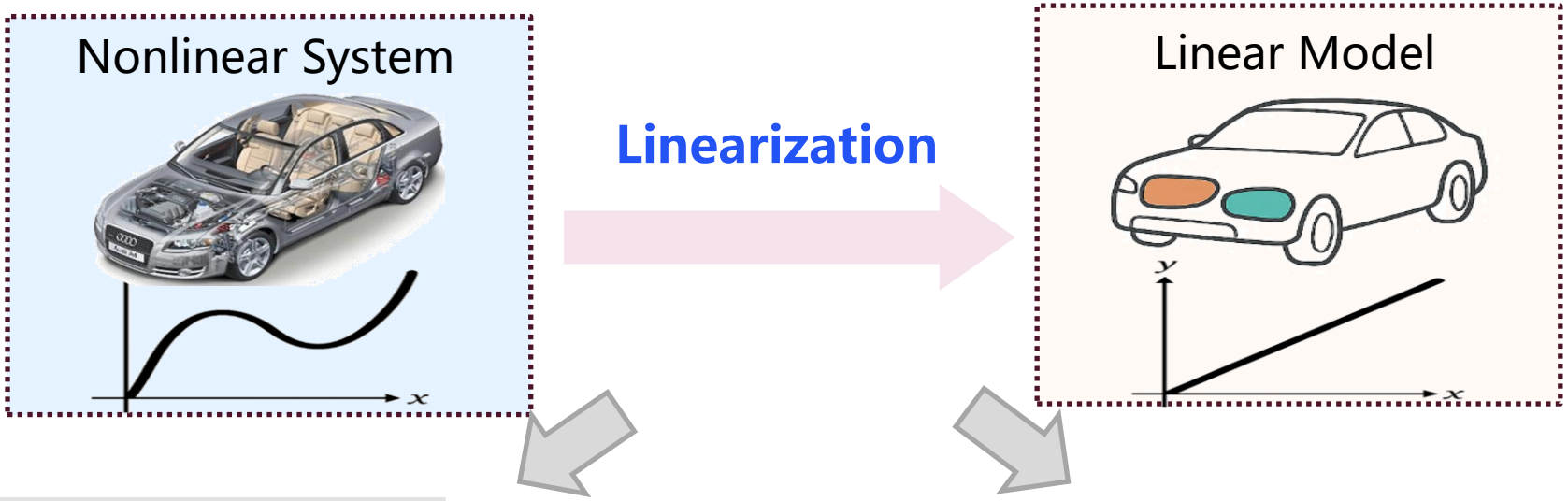
$$\min_{\hat{x}_t} \mathbb{E}\{(x_t - \hat{x}_t)^2\}$$

② Maximum A Posterior

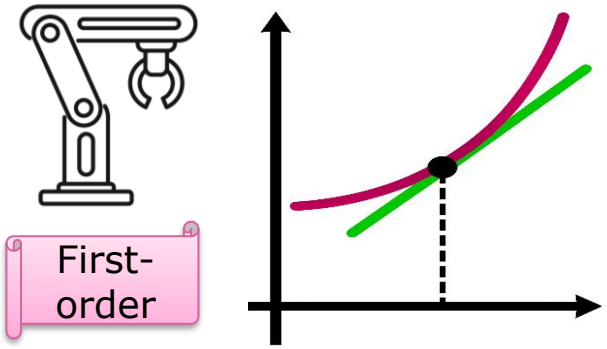
$$\max_{x_t} p(x_t|y_{1:t})$$

Challenges in Handling Nonlinearity

- **From Kalman filter to extended version or unscented version**



Taylor Series Expansion

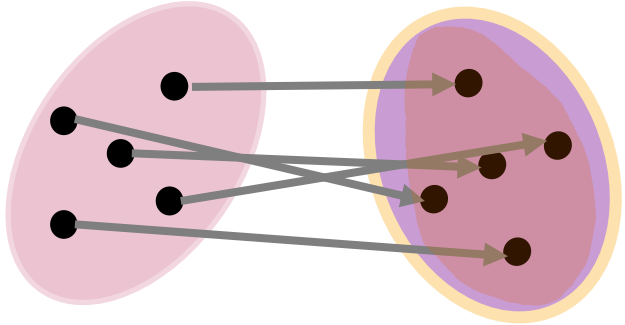


Extended Kalman Filter (EKF)

Suffer from large errors in strong nonlinear dynamics

$\hat{x} \longleftrightarrow x^*$

Unscented Transform



Unscented Kalman Filter (UKF)

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Optimization Viewpoint for Bayesian Filtering

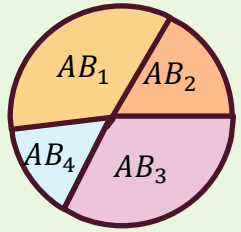
Variational problem for total probability rule

Prediction step:

$$p(A) = \int p(A|B)p(B)\,\mathrm{d}B$$

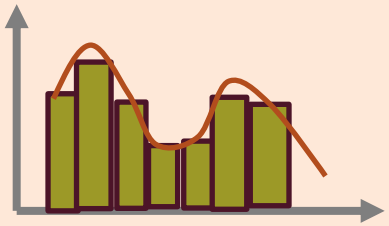
Classical view: aggregate over all possible conditions

$$p(A) = \int p(A, B)\mathrm{d}B$$



Optimization View: maximize log probability

$$p(A) = \underset{q}{\operatorname{argmax}} \left\{ \mathbb{E}_{p(B)} \mathbb{E}_{p(A|B)} \{\log q(A)\} \right\}$$



Optimization Viewpoint for Bayesian Filtering

Variational problem for Bayesian inference

Update step:

$$p(A|B) = \frac{p(A)p(B|A)}{\int p(B)p(B|A)\,\mathrm{d}A}$$

Classical view: compute the conditional probability

$$p(A|B) = \frac{p(A,B)}{p(B)}$$

parameter

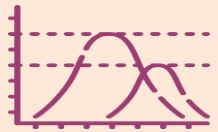
$$p(\text{parameter}|\text{data}) = \frac{p(\text{parameter},\text{data})}{p(\text{data})}$$

data

Optimization View: balance observation and prior

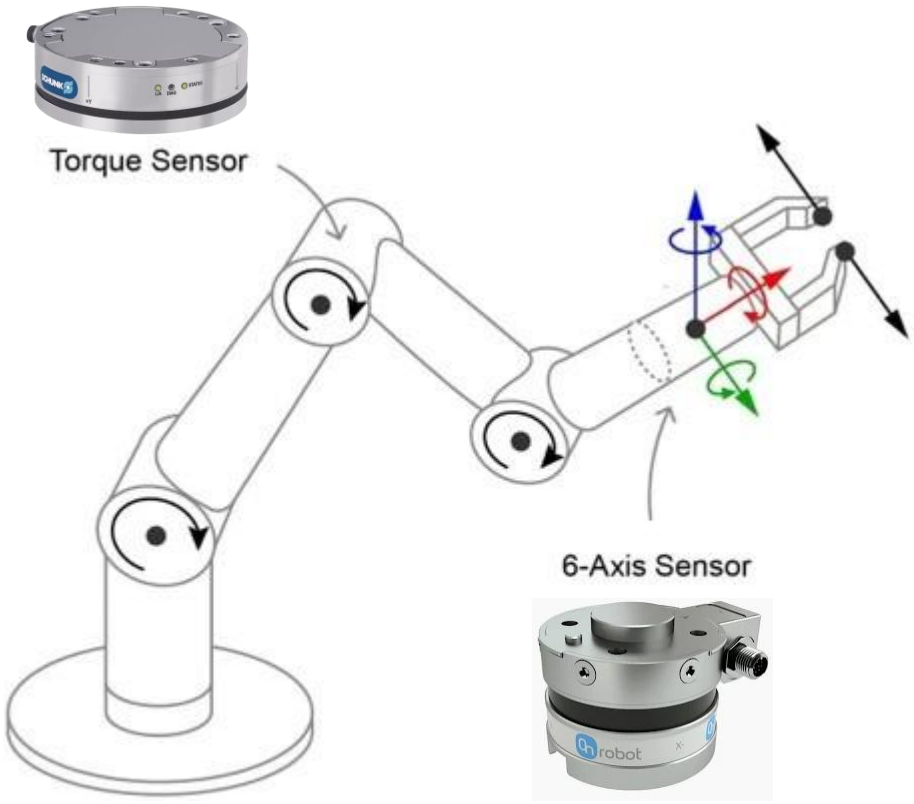
$$p(A|B) = \underset{q(A)}{\operatorname{argmin}}\left\{\mathbb{E}_{q(A)}\{-\log p(B|A)\} + D_{\mathrm{KL}}\big(q(A)||p(A)\big)\right\}$$



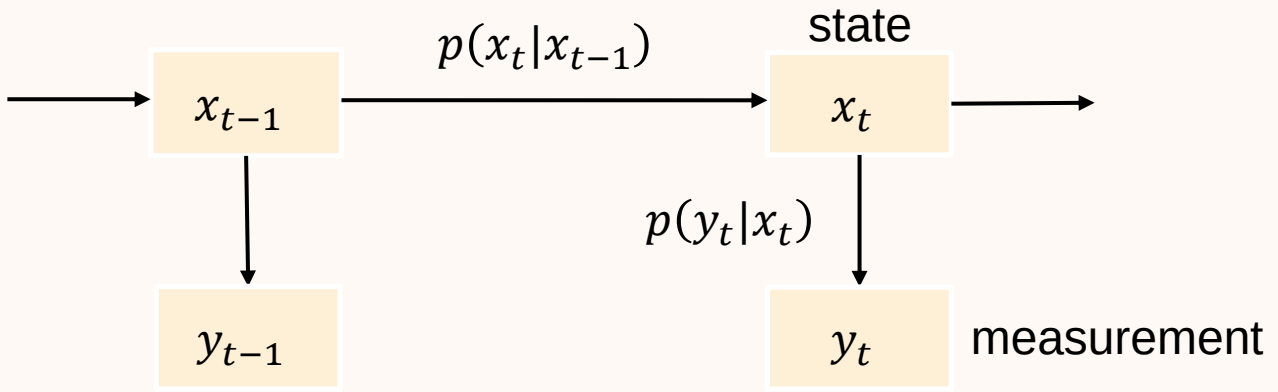


Stochastic Model for Nonlinear Systems

System description



① Hidden Markov Model (HMM)



② State Space Model (SSM)

Transition model

$$x_{t+1} = f(x_t) + \xi_t \qquad \mathbb{E}\{\xi_t\} = 0, \mathbb{D}\{\xi_t\} = Q$$

Measurement model

$$y_t = g(x_t) + \zeta_t \qquad \mathbb{E}\{\zeta_t\} = 0, \mathbb{D}\{\zeta_t\} = R$$

Variational Problems for Bayesian Filtering

Prediction Step

$$p(x_t|y_{1:t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|y_{1:t-1})\,\mathrm{d}x_{t-1}$$

Prediction Variational Problem

$$p(x_t|y_{1:t-1}) = \underset{q}{\operatorname{argmax}} \left\{ \mathbb{E}_{p(x_{t-1}|y_{1:t-1})} \mathbb{E}_{p(x_t|x_{t-1})} \underbrace{\{\log q(x_t)\}}_{\text{Maximum likelihood}} \right\}$$

Update Step

$$p(x_t|y_{1:t}) = \frac{p(x_t|y_{1:t-1})p(y_t|x_t)}{\int p(x_t|y_{1:t-1})p(y_t|x_t)\,\mathrm{d}x_t}$$

Update Variational Problem

$$p(x_t|y_{1:t}) = \underset{q}{\operatorname{argmin}} \left\{ \mathbb{E}_{q(x_t)} \underbrace{\{-\log p(y_t|x_t)\}}_{\text{Measurement Fitting}} + D_{\mathrm{KL}} \underbrace{\left(q(x_t)||p(x_t|y_{1:t-1})\right)}_{\text{Regularization}} \right\}$$

Variational Problems for Bayesian Filtering

Generalized form of Bayesian variational problems

Classic Bayesian

$$p(x_t|y_{1:t}) = \underset{q}{\operatorname{argmin}}\{\mathbb{E}_{q(x_t)}\{-\log p(y_t|x_t)\} + D_{\mathrm{KL}}(q(x_t)||p(x_t|y_{1:t-1}))\}$$

Measurement Fitting — Regularization

Generalization

Gibbs Loss $\ell(x_t, y_t)$
Design flexibility for better robustness and stability

Gibbs Bayesian (Generalized form)

$$p(x_t|y_{1:t}) = \underset{q}{\operatorname{argmin}}\{\mathbb{E}_{q(x_t)}\{\ell(x_t, y_t)\} + D_{\mathrm{KL}}(q(x_t)||p(x_t|y_{1:t-1}))\}$$

Measurement Fitting — Regularization

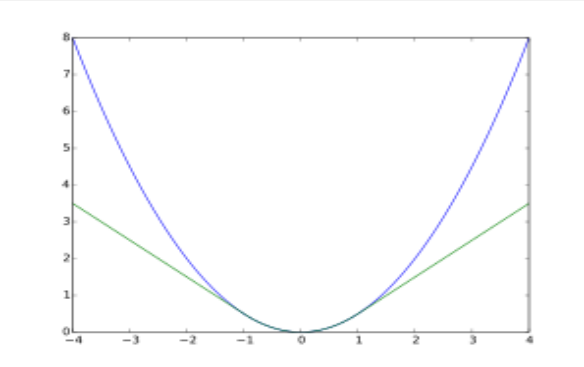
* Bissiri, Holmes, et al., *Journal of the Royal Statistical Society,* 2016

Variational Problems for Bayesian filtering

☐ Other choices of Gibbs loss function

① Composite Likelihood
e.g. Huber loss
Gandhi (2010)

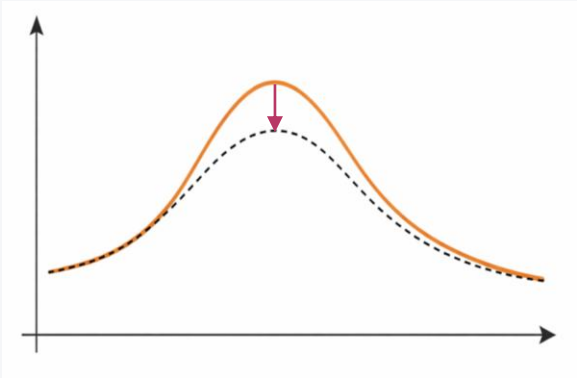
$$\ell(x_t, y_t) = \begin{cases} \underbrace{-|y_t - h(x_t)|^2 \quad \text{if } |y_t - h(x_t)| \le \delta}_{\text{Gaussian Likelihood}} \\ \underbrace{\delta\left(|y_t - h(x_t)| - \frac{1}{2}\delta\right) \quad \text{else}}_{\text{Laplace Likelihood}} \end{cases}$$



② Weighted Log-Likelihood
e.g. Convolutional loss
Cao (2024), Duran-Martin (2024)

$$\ell(x_t, y_t) = \underbrace{-w(x_t, y_t)}_{\text{Adjustable Parameter}} \log p(y_t|x_t)$$

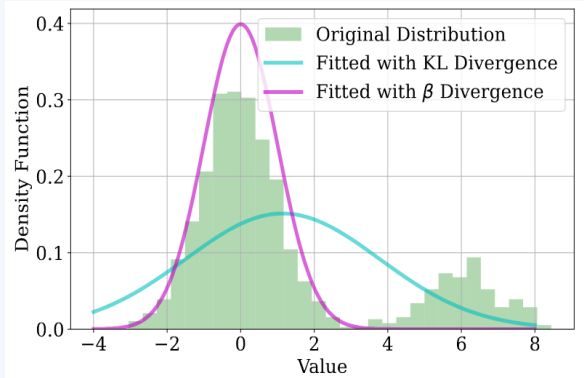
$$w(x_t, y_t) = \left(1 + \frac{\|y_t - g(x_t)\|^2_{R^{-1}}}{c^2}\right)^{-1}$$



③ Divergence-based Loss
e.g. Beta-cross entropy
Warwick (2020), Cao (2023)

$$\ell(x_t, y_t) = -\frac{\beta + 1}{\beta} p(y_t|x_t)^{\beta} + \int p(y|x_t)^{\beta+1}\, dy$$

Distribution adjustment

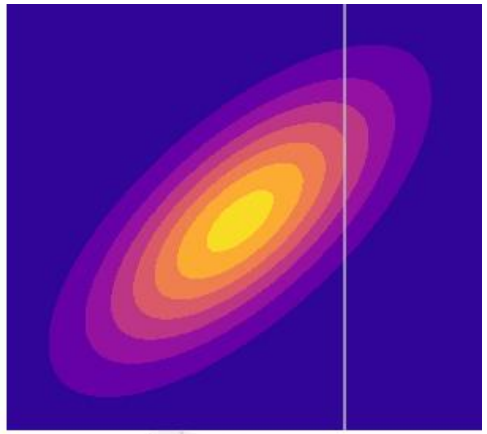


Extreme Conditions with Gaussian distribution

Prediction variational problem over Gaussian distribution

$L(\hat{x}_t, P_t)$ -Prediction Cost

$$\hat{x}_{t|t-1}, P_{t|t-1} = \underset{\hat{x}_t, P_t}{\operatorname{argmax}} \left\{ \mathbb{E}_{\mathcal{N}(\hat{x}_{t-1|t-1}, P_{t-1|t-1})} \mathbb{E}_{\mathcal{N}(f(x_{t-1}), Q)} \{ \log \mathcal{N}(\hat{x}_t, P_t) \} \right\}$$

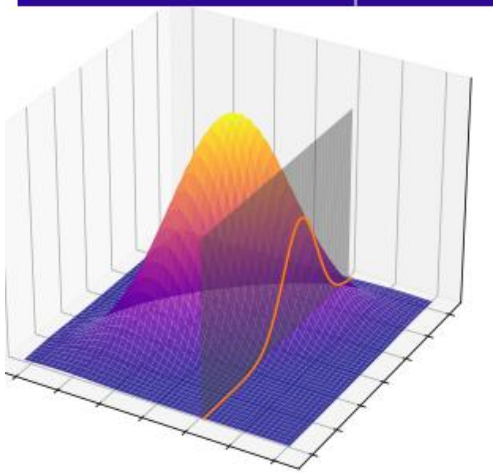


Stationary Point

Extreme condition

$$\hat{x}_{t|t-1} = \mathbb{E}_{\mathcal{N}(\hat{x}_{t-1|t-1}, P_{t-1|t-1})} \{ f(x_{t-1}) \}$$

$$P_{t|t-1} = \mathbb{E}_{\mathcal{N}(\hat{x}_{t-1|t-1}, P_{t-1|t-1})} \{ f(x_{t-1}) f^{\mathrm{T}}(x_{t-1}) \} + Q - \hat{x}_{t|t-1} \hat{x}_{t|t-1}^{\mathrm{T}}$$



The key is how to calculate expectation effectively!

Extreme Conditions with Gaussian distribution

- **Practical ways to calculate expectation with Moment Matching (MM) technique**

$$x \sim \mathcal{N}(\mu, P), P = LL^{\mathrm{T}}, \mathbb{E}\{f(x)\} \approx \sum_i W_i f(\mu + L\xi_i)$$

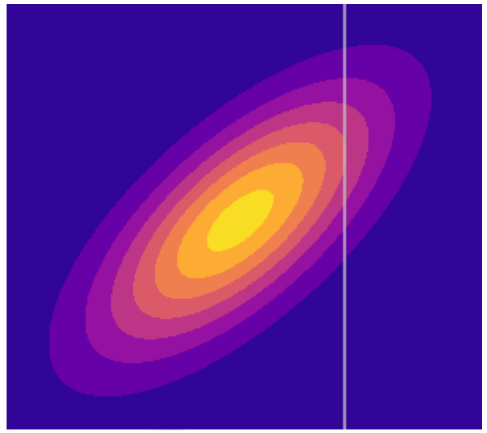
Method	Point set ξ_i	Weight W_i	#Num of Points	Property
Monte Carlo	$\xi_i \sim \mathcal{N}(0, I)$	$\frac{1}{N}$	N	High variance, slow convergence
Unscented transform	$\xi_0 = 0, \xi_i = \pm\sqrt{n+\lambda}e_i$	$W_0 = \frac{\lambda}{n+\lambda}$, $W_i = \frac{1}{2(n+\lambda)}$	$2n+1$	Not scalable in high dim
Cubature rule	$\xi_i = \pm\sqrt{n}e_i$	$W_i = \frac{1}{2n}$	$2n$	Not scalable in high dim
Gauss-Hermite transform	$\xi_i \in [z_1, z_2, \dots, z_m]^{\times n}$ $H_m(z_j) = 0$	$W_i = \prod_{k=1}^{n} \frac{2^{m-1} m! \sqrt{\pi}}{m^2 [H_m(z_k)]^2}$	m^n	Exponential complexity

Extreme Conditions with Gaussian distribution

Update variational problem over Gaussian distribution

$J(\hat{x}_t, P_t)$ - Update Cost

$$\hat{x}_{t|t}, P_{t|t} = \underset{\hat{x}_t, P_t}{\operatorname{argmin}} \left\{ \mathbb{E}_{\mathcal{N}(\hat{x}_t, P_t)}\{\ell(x_t, y_t)\} + D_{\mathrm{KL}}\left(\mathcal{N}(x_t; \hat{x}_t, P_t) || \mathcal{N}\left(\hat{x}_{t|t-1}, P_{t|t-1}\right)\right)\right\}$$



Stationary Point

$$\frac{\partial J(x_t, P_t)}{\hat{x}_t} = \mathbb{E}_{\mathcal{N}(x_t; \hat{x}_t, P_t)}\{\ell'(x_t, y_t)\} + P_{t|t-1}^{-1}\left(\hat{x}_t - \hat{x}_{t|t-1}\right) = 0$$

$$\frac{\partial J(x_t, P_t)}{P_t^{-1}} = -\frac{1}{2} P_t \mathbb{E}_{\mathcal{N}(x_t; \hat{x}_t, P_t)}\{\ell''(x_t, y_t)\} P_t - \frac{1}{2} P_t P_{t|t-1}^{-1} P_t + \frac{1}{2} P_t = 0$$

Stein's Lemma

$$\frac{\partial \mathbb{E}\{\ell\}}{\partial \hat{x}} = \mathbb{E}\{\ell'\}$$

$$\frac{\partial \mathbb{E}\{\ell\}}{\partial P^{-1}} = -\frac{1}{2} P \mathbb{E}\{\ell''\} P$$



Charles Stein

T. Barfoot (2024) State estimation for robotics. Cambridge University Press

Extreme Conditions with Gaussian distribution

Update variational problem over Gaussian distribution

Stationary point for each update step

$$\mathbb{E}_{\mathcal{N}(x_t;\hat{x}_{t|t},P_{t|t})}\{\ell'(x_t,y_t)\}+P_{t|t-1}^{-1}(\hat{x}_{t|t}-\hat{x}_{t|t-1})=0$$

$$-\frac{1}{2}P_{t|t}\mathbb{E}_{\mathcal{N}(x_t;\hat{x}_{t|t},P_{t|t})}\{\ell''(x_t,y_t)\}P_{t|t}-\frac{1}{2}P_{t|t}P_{t|t-1}^{-1}P_{t|t}+\frac{1}{2}P_{t|t}=0$$

Left Multiply $P_{t|t-1}$

Left & Right Multiply $P_{t|t}^{-1}$

Extreme condition

$$\hat{x}_{t|t}=\hat{x}_{t|t-1}-P_{t|t-1}\mathbb{E}_{\mathcal{N}(\hat{x}_{t|t},P_{t|t})}\{\ell'(x_t,y_t)\}$$

$$P_{t|t}^{-1}=P_{t|t-1}^{-1}+\mathbb{E}_{\mathcal{N}(\hat{x}_{t|t},P_{t|t})}\{\ell''(x_t,y_t)\}$$

Both are implicit nonlinear equations and $\hat{x}_{t|t}$ and $P_{t|t}^{-1}$ are tightly coupled!

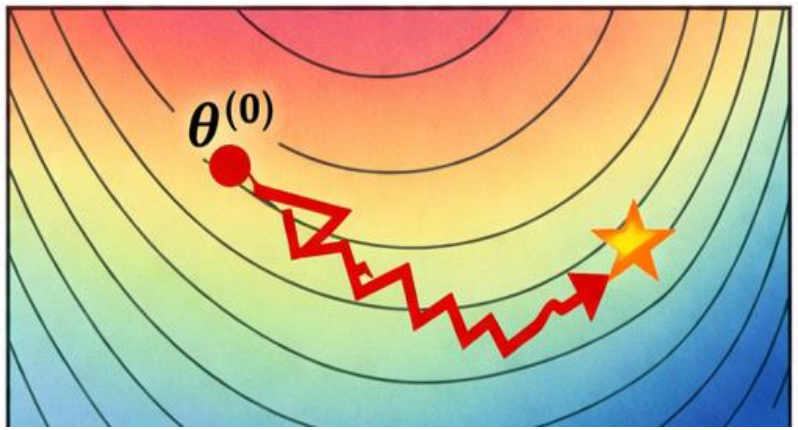
Iterative Optimization of Bayesian Variational Problems

Iterative optimization using first-order derivative

$$\min_{\hat{x}_t, P_t} J(\hat{x}_t, P_t) \quad \text{-- Update cost}$$

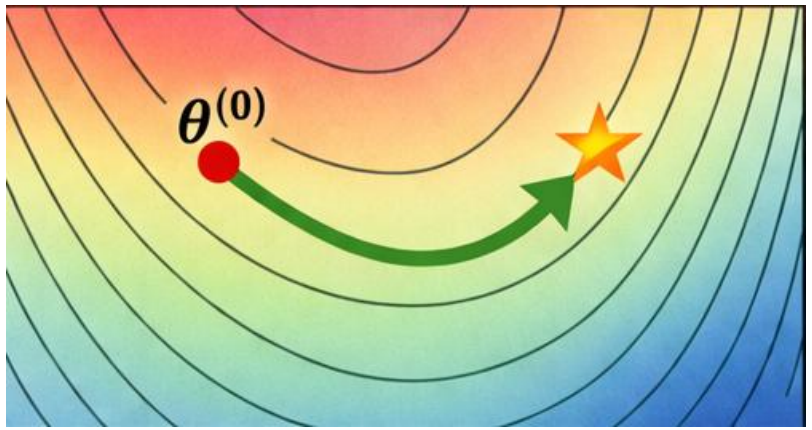
① Gradient Descent

- Slow convergence
- Zigzagging path when ill-conditioned



② Natural Gradient Descent

- Fast convergence
- Smooth path in probability manifold



Any other derivative-based optimization

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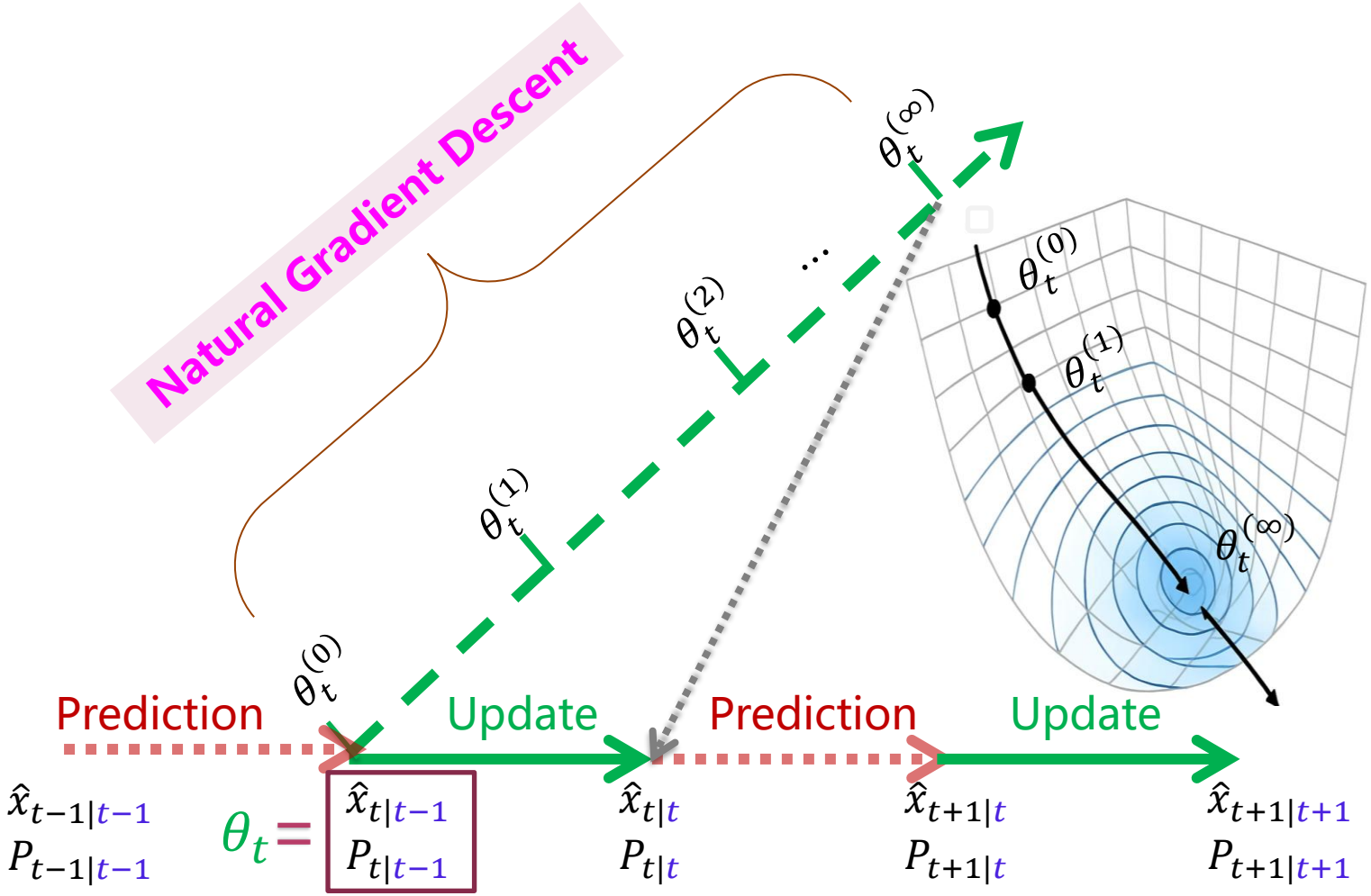
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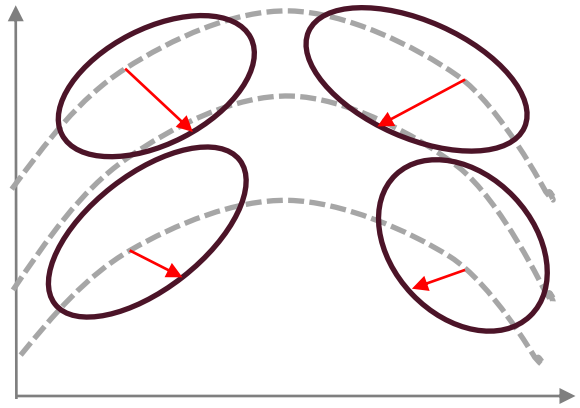
4 Experimental Results

Design of Geometry-aware Filter

- **Key idea: Use iterative optimization to replace extreme condition**



Natural Gradient



Iterative fomula:

$$\theta^{(i+1)} = \theta^{(i)} - F_{\text{fisher}}^{-1} \frac{\partial J}{\partial \theta}$$

Design of Geometry-aware Filter

Natural gradient for Gaussian manifold

Gibbs Loss

$$J = \mathbb{E}_{\mathcal{N}(\hat{x}_t, P_t)}\{\ell(x_t, y_t)\} + D_{\mathrm{KL}}\left(\mathcal{N}(\hat{x}_t, P_t) || \mathcal{N}\left(\hat{x}_{t|t-1}, P_{t|t-1}\right)\right)$$

Update Cost

Normal Gradient

Natural gradient descent

$$\theta^{(i+1)} = \theta^{(i)} - F_{\text{fisher}}^{-1} \frac{\partial J}{\partial \theta}$$

$$\frac{\partial J(x_t, P_t)}{\hat{x}_t} = \mathbb{E}_{\mathcal{N}(x_t; \hat{x}_{t|t}, P_{t|t})}\{\ell'(x_t, y_t)\} + P_{t|t-1}^{-1}\left(x_t - \hat{x}_{t|t-1}\right)$$

$$\frac{\partial J(x_t, P_t)}{P_t^{-1}} = -\frac{1}{2} P_t \mathbb{E}_{\mathcal{N}(x_t; \hat{x}_{t|t}, P_{t|t})}\{\ell''(x_t, y_t)\} P_t - \frac{1}{2} P_t P_{t|t-1}^{-1} P_t + \frac{1}{2} P_t$$

$$\theta = [\hat{x}_t, \mathrm{vec}(P_t^{-1})]^{\top}$$

Design of Geometry-aware Filter

Natural gradient for Gaussian manifold

$$F_{\text{fisher}} \triangleq \mathbb{E}\left\{\left[\frac{\log \mathcal{N}(\hat{x}_t;\ P_t)}{\partial \theta}\right]\left[\frac{\log \mathcal{N}(\hat{x}_t;\ P_t)}{\partial \theta}\right]^{\top}\right\}$$

Fisher Matrix

Inverse of Fisher matrix

Natural gradient descent

$$\theta^{(i+1)} = \theta^{(i)} - F_{\text{fisher}}^{-1}\frac{\partial J}{\partial \theta}$$

$$F_{\text{fisher}}^{-1} = \begin{bmatrix} P_t & 0 \\ 0 & 2(P_t^{-1} \otimes P_t^{-1}) \end{bmatrix}$$

$$\theta = [\hat{x}_t, \text{vec}(P_t^{-1})]^{\top}$$

P_t^{-1}: Precision matrix VS P^t : Covariance matrix

Two Kinds of Explict Iterative Formula

Explicit iterative formula of the first kind

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\{\ell''(x_t, y_t)\}$$

$$\hat{x}_t{}^{(i+1)} = \hat{x}_t{}^{(i)} - P_t^{(i+1)} \cdot \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\{\ell'(x_t, y_t)\} - P_t^{(i+1)} \cdot P_{t|t-1}^{-1}\left(x_t^{(i)} - \hat{x}_{t|t-1}\right)$$

(First/Second-Order Derivatives of Gibbs Loss)

Use moment matching (MM) to calculate expectation

$$\mathbb{E}_{\mathcal{N}(x_t;\hat{x},P)}\{f(\cdot)\} \cong \mathrm{MM}\left(\hat{x}, P; f(\cdot)\right)$$

High computational efficiency but may suffer from non-smoothing Gibbs loss



Two Kinds of Explict Iterative Formula

Explicit iterative formula of the second kind

Use Stein's Lemma again!

$$\mathbb{E}\{\ell'\} = P^{-1}\,\mathbb{E}\{(x-\hat{x})\ell\}$$

$$\mathbb{E}\{\ell''\} = P^{-1}\,\mathbb{E}\{(x-\hat{x})(x-\hat{x})^{\mathrm{T}}\ell\}P^{-1} - P^{-1}\,\mathbb{E}\{\ell\}$$

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + (P_t^{-1})^{(i)}\mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\left\{\left(x_t-\hat{x}_t^{(i)}\right)\left(x_t-\hat{x}_t^{(i)}\right)^{\mathrm{T}}\ell(x_t,y_t)\right\}(P_t^{-1})^{(i)} - (P_t^{-1})^{(i)}\mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\{\ell(x_t,y_t)\}$$

$$\hat{x}_t{}^{(i+1)} = \hat{x}_t{}^{(i)} - P_t^{(i+1)}(P_t^{-1})^{(i)}\mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\left\{\left(x_t-\hat{x}_t^{(i)}\right)\ell(x_t,y_t)\right\} - P_t^{(i+1)}\cdot P_{t|t-1}^{-1}\left(x_t^{(i)}-\hat{x}_{t|t-1}\right)$$

(Only Zero-Order Information of Gibbs Loss)

Need to calculate one more expectation, thus slightly increasing computational burden; but less affected by non-smoothness of Gibbs loss

Geometric Explanation of Natural Gradient Updates

Information geometry: Natural Gradient = Steepest Descent on Gaussian Manifold

Update Objective

$$\min_{p(x_t|y_{1:t})} J(\hat{x}_t, P_t)$$

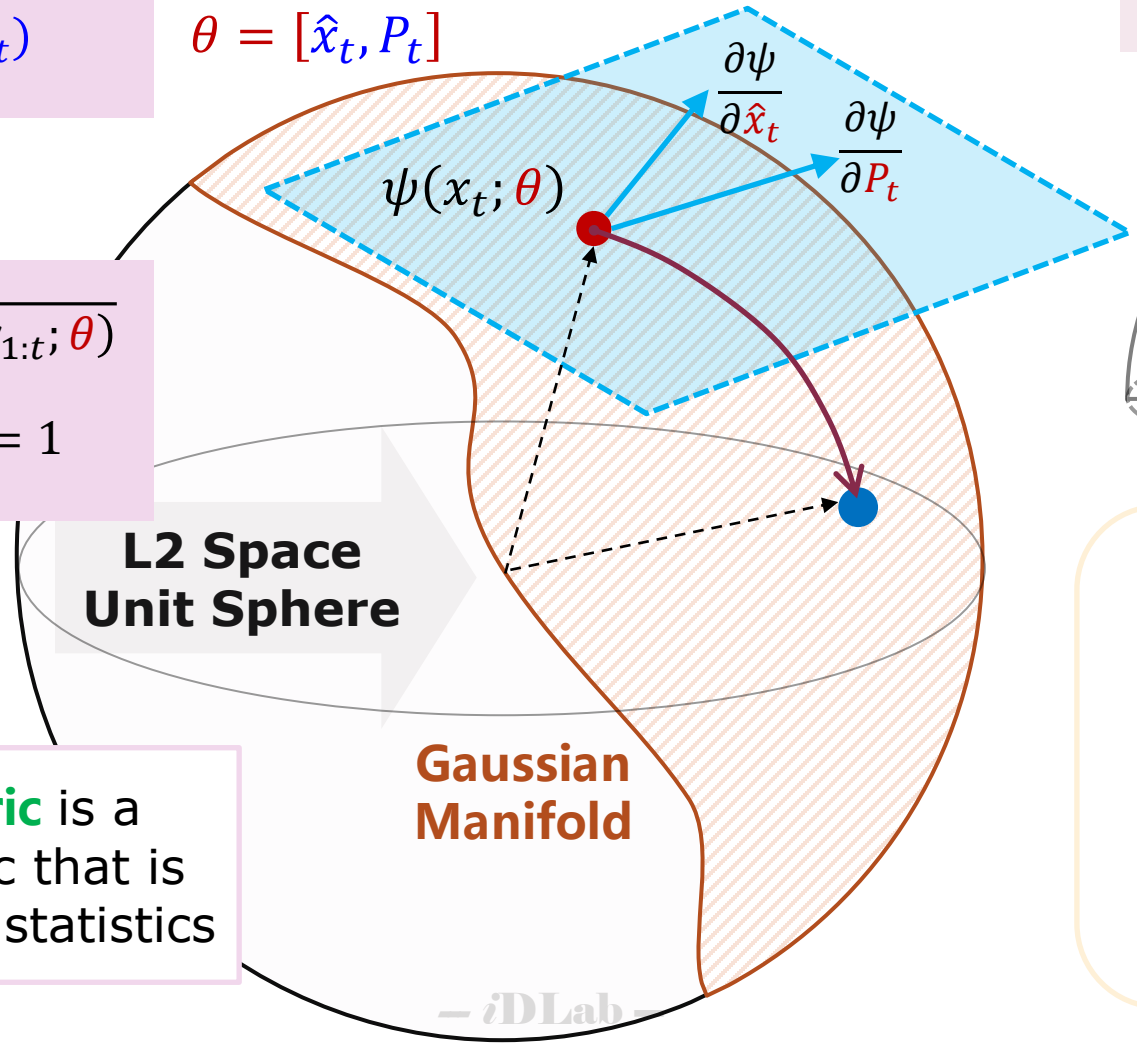
Subj. to

Probabilistic Constraints

$$\psi(x_t;\theta) = \sqrt{p(x_t|y_{1:t};\theta)}$$

$$\int \psi(x_t;\theta)^2 \, \mathrm{d}x_t = 1$$

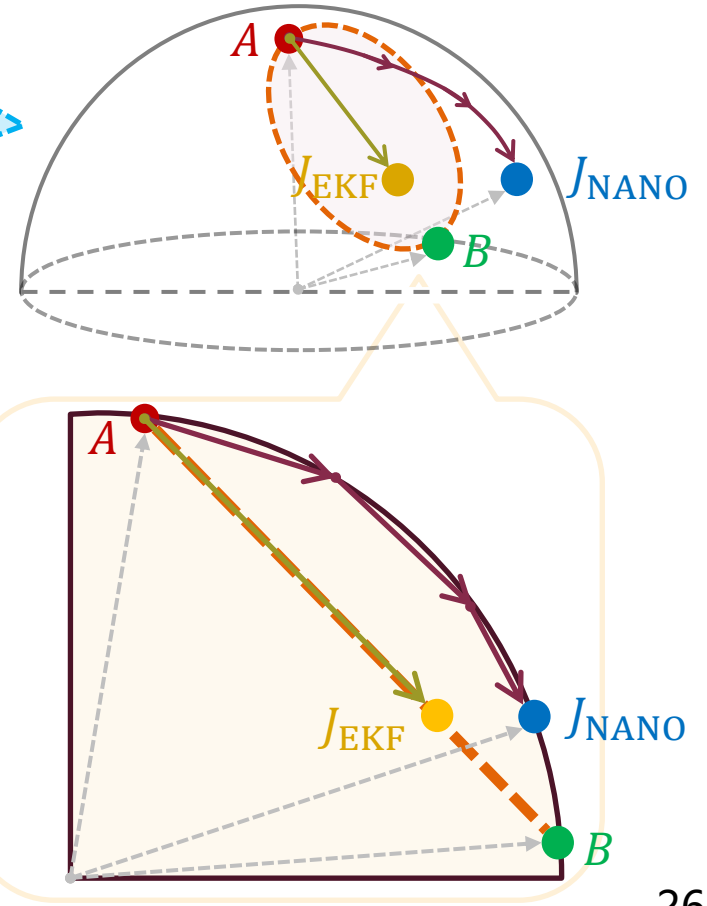
$$\theta = [\hat{x}_t, P_t]$$



Chentsov Theorem

Fisher information metric is a unique Riemannian metric that is **invariant** under sufficient statistics

EKF vs NANO



– iDLab –

Natural grAdient gaussiaN apprOximation (NANO) Filter

Pseudo-code of NANO (w/ iterative formula of first kind)

- (1) Prediction Step
  - (1-1) Calculate prior mean: $\{\hat{x}_{t|t-1}, W\} = \mathrm{MM}\left(\hat{x}_{t-1|t-1}, P_{t-1|t-1}; f(\cdot, u_t)\right)$
  - (1-2) Calculate prior covariance: $P_{t|t-1} = W + Q$
- (2) Update Step
  - (2-1) Initialize posterior mean $\hat{x}_t^{(0)}$ and posterior covariance $P_t^{(0)}$
  - (2-2) For k, do iteration
    - (i) Define Gibbs loss function $\ell(x, y_t) = \frac{1}{2}\|y_t - g(x)\|_{R^{-1}}^2$

Other options:
① Huber loss
② Convolutional loss
③ Beta-cross entropy

    - (ii) Calculate expected coefficients

$$\left\{V_x^{(k)}, \cdot\right\} = \mathrm{MM}\left(\hat{x}_t^{(k)}, P_t^{(k)}; \frac{\partial \ell(x, y_t)}{\partial x}\right), \left\{V_{xx}^{(k)}, \cdot\right\} = \mathrm{MM}\left(\hat{x}_t^{(k)}, P_t^{(k)}; \frac{\partial^2 \ell(x, y_t)}{\partial x^2}\right)$$

    - (iii) Calculate intermediate mean and covariance

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + V_{xx}^{(k)}$$

$$\hat{x}_t{}^{(i+1)} = \hat{x}_t{}^{(i)} - P_t^{(i+1)} \cdot V_x^{(k)} - P_t^{(i+1)} \cdot P_{t|t-1}^{-1}\left(x_t^{(i)} - \hat{x}_{t|t-1}\right)$$

The step size can be added to adjust NANO's convergence speed

  - (2-3) Update posterior mean and posterior covariance $\hat{x}_{t|t} = \hat{x}_t^{(k+1)}, P_{t|t} = P_t^{(k+1)}$

Precision Matrix vs Covariance Matrix

- **Different forms of update formula in NANO**

Precision matrix

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\{\ell''(x_t, y_t)\}$$

Cao, Zhang, et al., arxiv, 2024

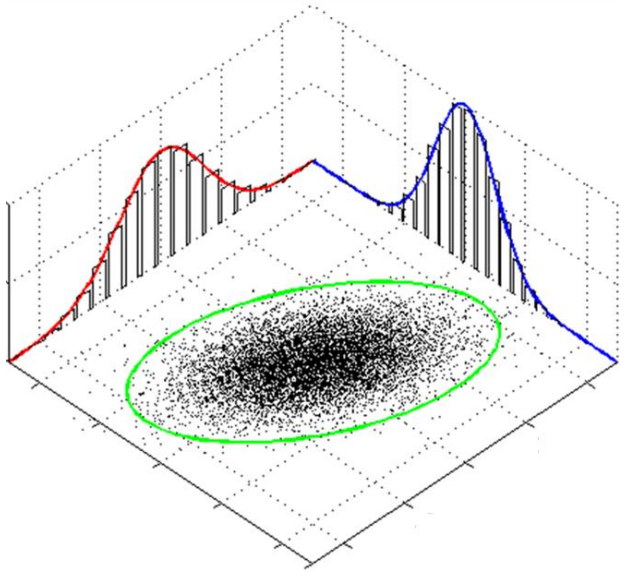
Covariance matrix

$$P_t^{(i+1)} = P_t^{(i)} - P_t^{(i)}\left(P_{t|t-1}^{-1} - \left(P_t^{(i)}\right)^{-1} - \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\{\ell''(x_t, y_t)\}\right)P_t^{(i)}$$

Li, Liu, et al., ICIF, 2025

NANO with Positive Definiteness (NANO-P)

Covariance Matrix Need to be Positive Definite (PD)



Challenge in Covariance Update

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\left\{\frac{\partial^2 \ell(x_t,y_t)}{\partial x_t^{\mathrm{T}} \partial x_t}\right\}$$

PD

PD or non-PD ?

Why NANO's update fails to be PD

Measurement Model

$$y_t = h(x_t) + \zeta_t$$

Gibbs Loss Function

$$\ell(x_t,y_t) = \frac{1}{2}\|y_t - h(x_t)\|_{R^{-1}}^2$$

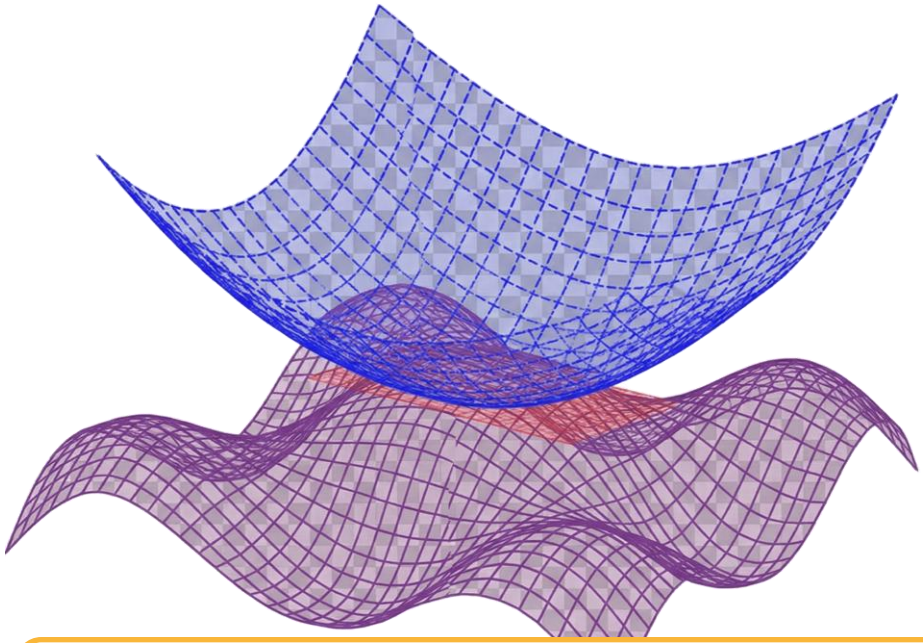
$$\frac{\partial^2 \ell(x_t,y_t)}{\partial x_t^{\mathrm{T}} \partial x_t} = \underbrace{\frac{\partial h}{\partial x_t^{\mathrm{T}}} R^{-1} \frac{\partial h}{\partial x_t}}_{\text{First term: Semi-PD}} - \underbrace{\sum_{j=1}^{m} [R^{-1}(y_t - h(x_t))]_j \frac{\partial^2 h_j}{\partial x_t^2}}_{\text{Second term: Non-PD}}$$

NANO with Positive Definiteness (NANO-P)

Method I: Hessian Approximation

$$\frac{\partial^2 \ell(x_t, y_t)}{\partial x_t^{\mathrm{T}} \partial x_t} \approx \frac{\partial h}{\partial x_t^{\mathrm{T}}} R^{-1} \frac{\partial h}{\partial x_t}$$

Only keep the first term !



Equivalent to quadratic approximation of Gibbs loss $\ell(x)$

Why Hessian Approximation Works?

Let define $r(x) = \sqrt{R}\big(y - h(x)\big)$, then $\ell(x) = \frac{1}{2}\|r(x)\|_2^2$

(1) $$\ell(x + \Delta x) \approx \ell(x) + \frac{\partial \ell^{\mathrm{T}}}{\partial x} \Delta x + \frac{1}{2} \Delta x^{\mathrm{T}} \frac{\partial^2 \ell}{\partial x^{\mathrm{T}} \partial x} \Delta x$$

$$=$$

(2) $$\|r(x + \Delta x)\|_2^2 \approx \underbrace{\left\| r(x) + \frac{\partial r}{\partial x} \Delta x \right\|_2^2}_{\text{Linearization}} = 2\ell(x) + 2r(x)^{\mathrm{T}} \frac{\partial r}{\partial x} \Delta x + \Delta x^{\mathrm{T}} \frac{\partial r^{\mathrm{T}}}{\partial x} \frac{\partial r}{\partial x} \Delta x$$

NANO with Positive Definiteness (NANO-P)

Method II: Cholesky Decomposition

$$P_t^{-1} = \Sigma_t \Sigma_t^{\top}$$

Use Σ_t to replace P_t^{-1} !

- Iterative formula of the first kind

$$(P_t^{-1})^{(i+1)} = P_{t|t-1}^{-1} + \mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\left\{\frac{\partial^2 \ell(x_t,y_t)}{\partial x_t^{\top}\partial x_t}\right\}$$

- Iterative formula with Cholesky Factor

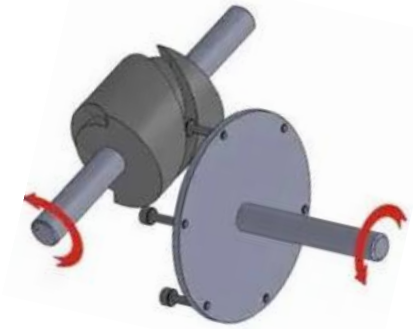
Reparameterization $P_t^{-1} = \Sigma_t \Sigma_t^{\top}$

$$\Sigma_t^{(i+1)} = \Sigma_t^{(i)} \exp\left(\frac{1}{2}(\Sigma_t^{(i)})^{-1}(\mathbb{E}_{\mathcal{N}\left(x_t;\hat{x}_t^{(i)},P_t^{(i)}\right)}\left\{\frac{\partial^2 \ell(x_t,y_t)}{\partial x_t^{\top}\partial x_t}\right\} + P_{t|t-1}^{-1})(\Sigma_t^{(i)})^{-\top}\right)$$

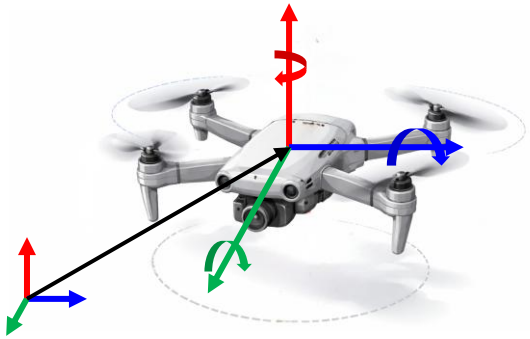
$$(P_t^{-1})^{(i+1)} = \Sigma_t^{(i+1)}(\Sigma_t^{\top})^{(i+1)}$$

NANO on Lie Group (NANO-L)

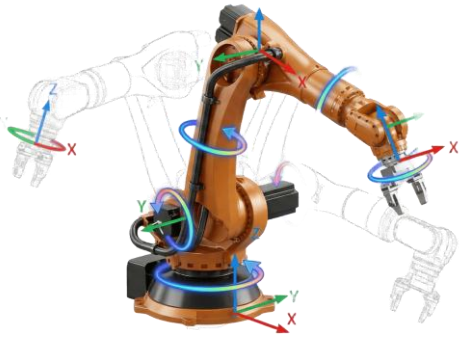
Mathematical model using Lie group



Rotated Parts



6DOF Drone



Articulated Arm

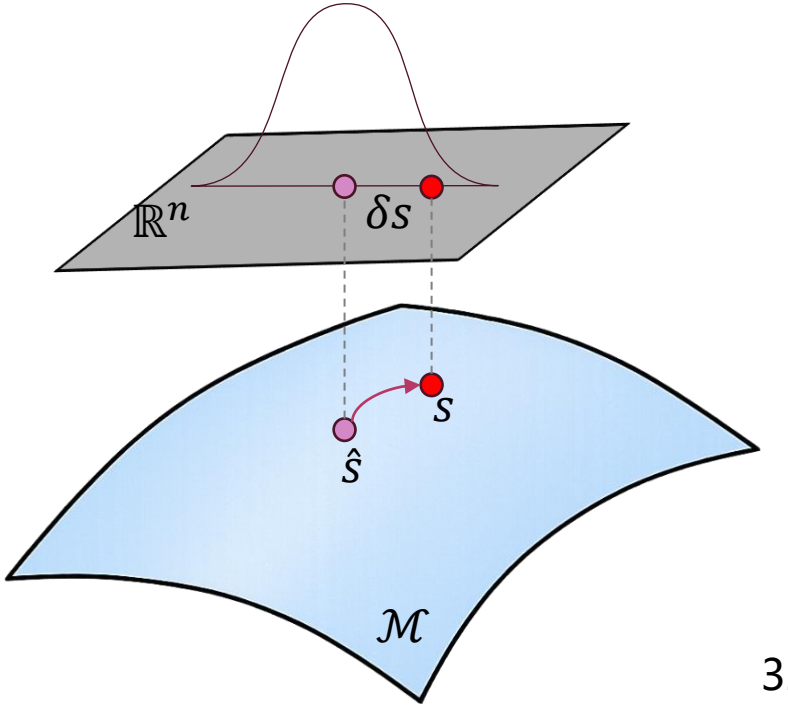
How to define states with Lie group

Tangent Space Decomposition

$$s = \hat{s} \boxplus \delta s$$

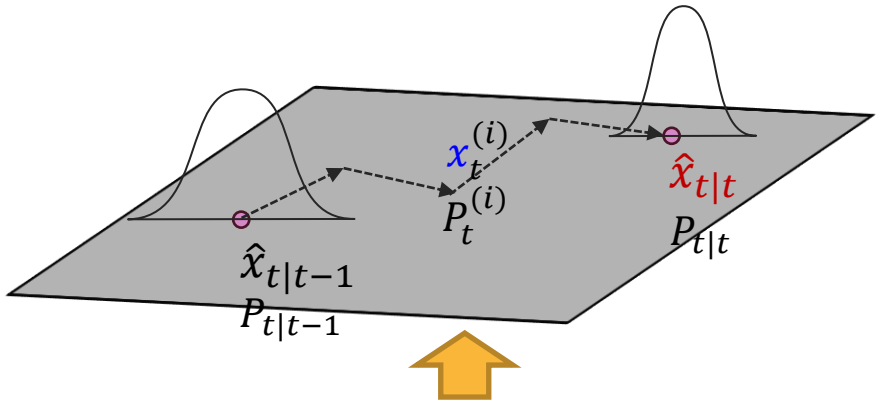
State in Lie group = Nominal state in Lie group + Error state in tangent space

$$\hat{s} \in \mathcal{M}, \delta s \in \mathbb{R}^n \Rightarrow \hat{s} \boxplus \delta s \in \mathcal{M}$$



NANO on Lie Group (NANO-L)

$$\hat{x}_{t|t} = x_t^{(\infty)}, P_{t|t} = P_t^{(\infty)}$$

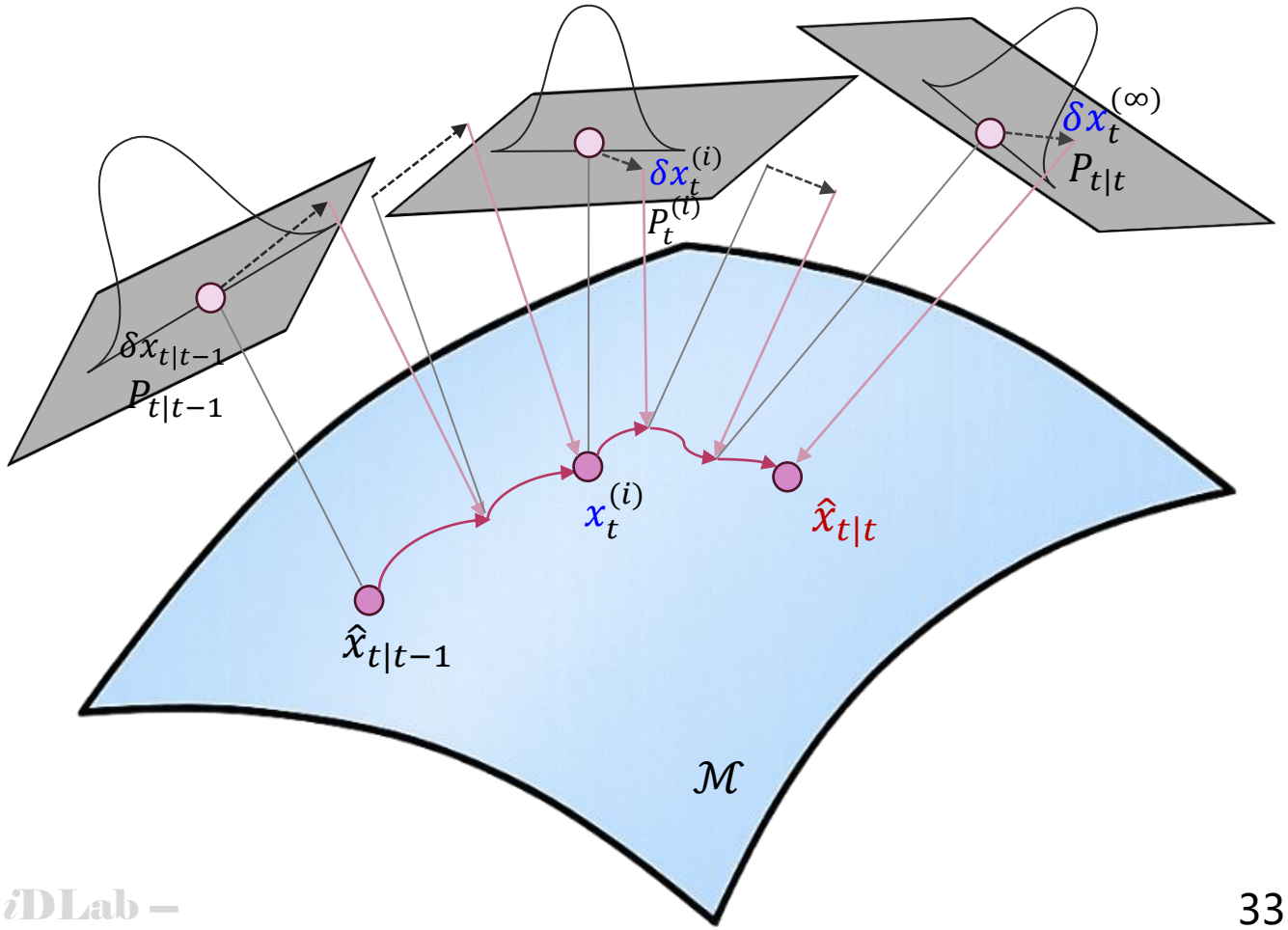


Gaussian iteration in Euclidian space

VS

Gaussian iteration in tangent space of Lie group

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} \boxplus \sum_{i=0}^{\infty} \delta x_t^{(i)}, P_{t|t} = P_t^{(\infty)}$$



Contents

1 Motivation

2 Variational Problem for Bayesian Filtering

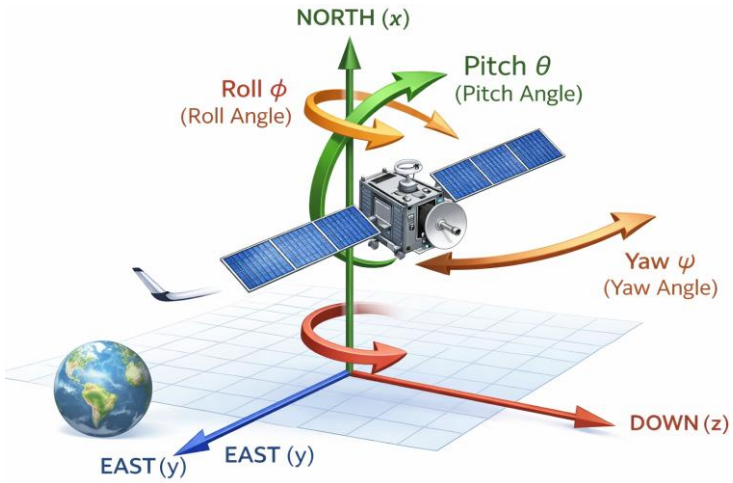
3 NANO Filtering Algorithm Design

4 Experimental Results

Task 1: Earth Orbit Satellite

Satellite attitude estimation

- State $\theta \in \mathbb{R}^3$, including pitch, roll and yaw angle
- Measurement $y \in \mathbb{R}^6$, including gravitational and geomagnetic sensors
- Measurement model is nonlinear



Sinusoidal angular velocity

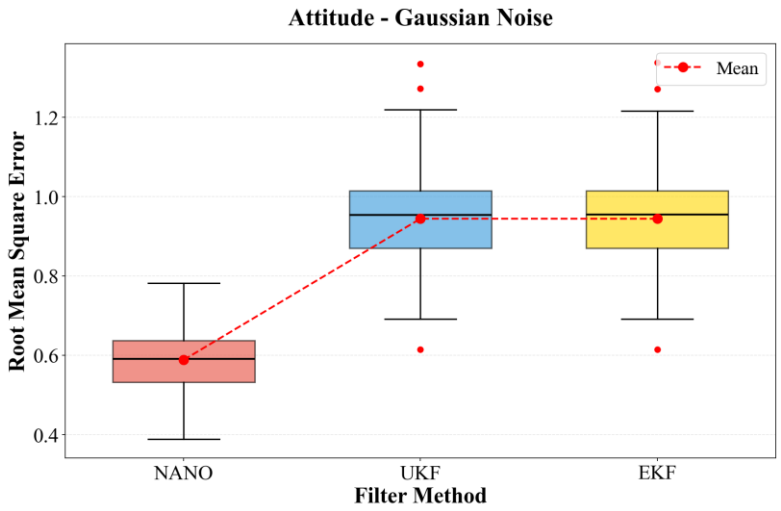
$$\frac{\pi}{18}\sin(2f\pi t)\cdot I_{3\times 1}$$

$\hat{x}_0$	Method	RMSE	Time [ms]
$[0\,0\,0]^T$	EKF	0.944 (0.020)	0.251
	UKF	0.944 (0.020)	0.387
	NANO	0.589 (0.006)	3.084
$[\frac{\pi}{18}\ \frac{\pi}{18}\ \frac{\pi}{18}]^T$	EKF	1.576 (0.017)	0.268
	UKF	1.576 (0.017)	0.429
	NANO	1.390 (0.018)	3.189

Constant angular velocity

$$\frac{\pi}{18\sqrt{2}}\cdot I_{3\times 1}$$

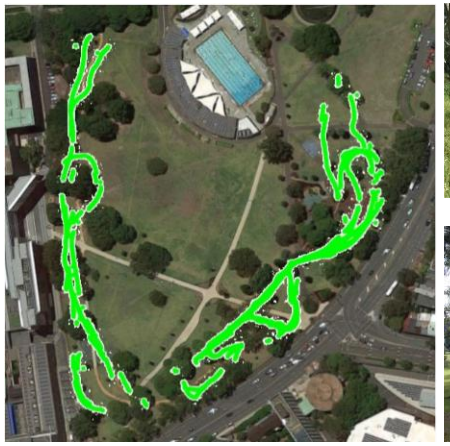
$\hat{x}_0$	Method	RMSE	Time [ms]
$[0\,0\,0]^T$	EKF	0.971 (0.021)	0.262
	UKF	0.971 (0.021)	0.415
	NANO	0.597 (0.007)	2.946
$[\frac{\pi}{18}\ \frac{\pi}{18}\ \frac{\pi}{18}]^T$	EKF	1.592 (0.018)	0.248
	UKF	1.592 (0.018)	0.383
	NANO	1.391 (0.018)	3.093



Task 2: Off-road Vehicle SLAM

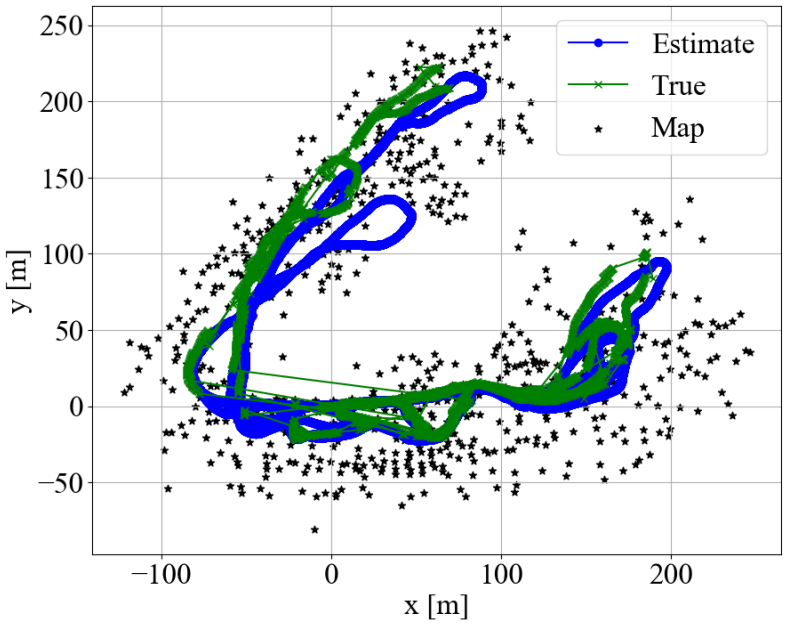
Victoria park dataset

- State $x \in \mathbb{R}^3$, including vehicle's 2D position and heading angle
- Measurement $y \in \mathbb{R}^{2N}$, including N range-bearing measurements
- State transition and measurement model are nonlinear models

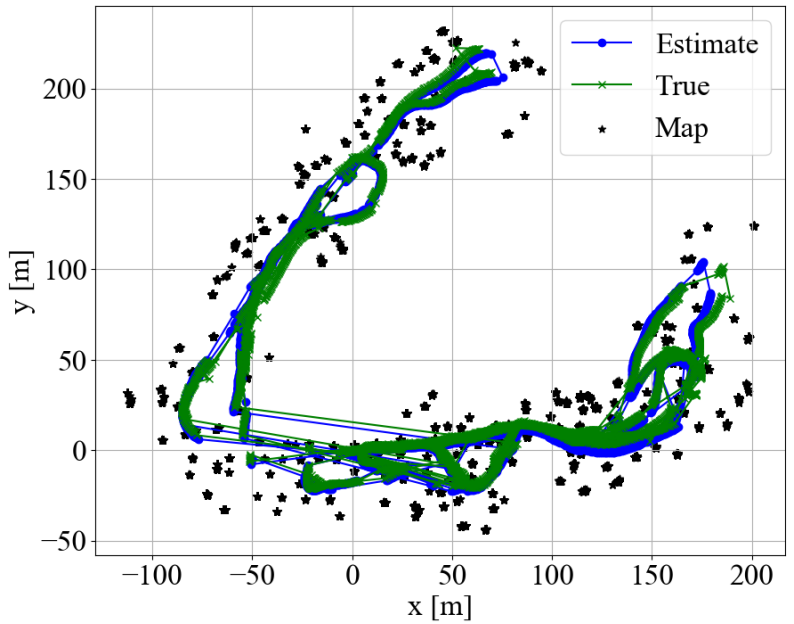








Case 1:EKF-SLAM



Case 2:NANO-SLAM (Ours)

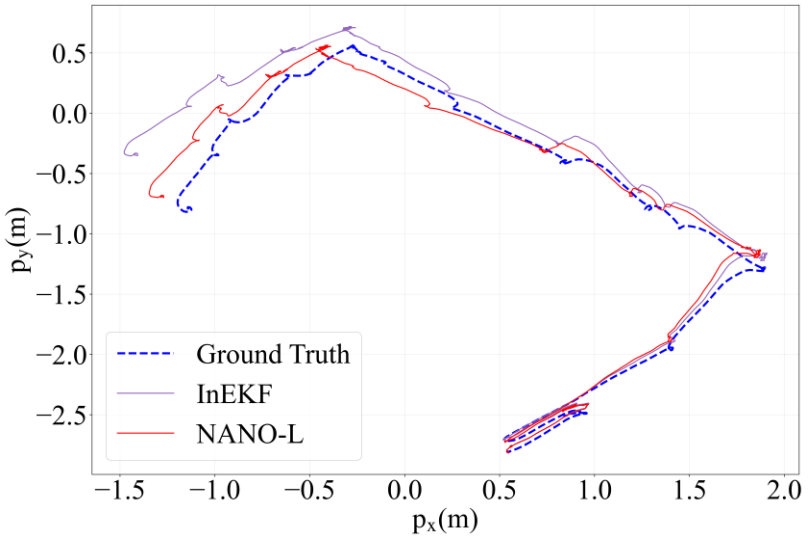
Method	RMSE [m]	Time [s]
EKF-SLAM	7.783	0.312
Ufast-SLAM	5.147	0.248
NANO-SLAM	**2.538**	**0.270**

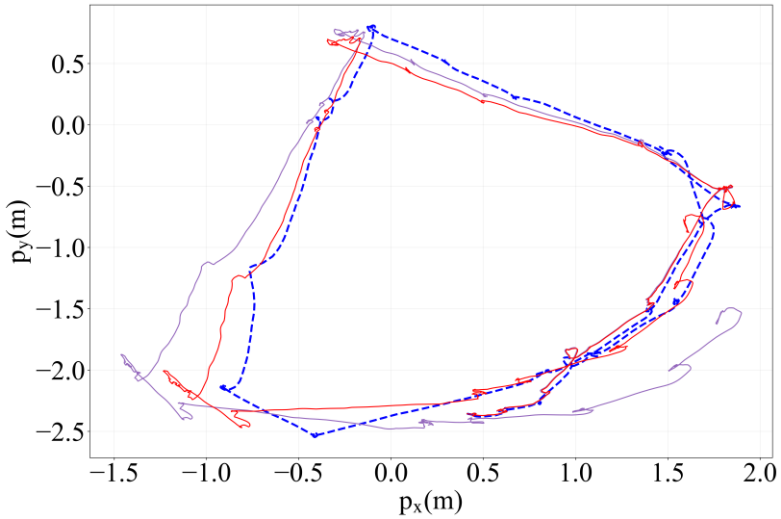
Task 3: Quadruped Robot Pose Estimation

Unitree Go2 robot dog

- State $x \in \mathbb{R}^{27}$, including robot's pose, velocity, IMU bias, foots position
- Measurement $y \in \mathbb{R}^{12}$, including 4 foot-body positions
- Both transition and measurement model are nonlinear





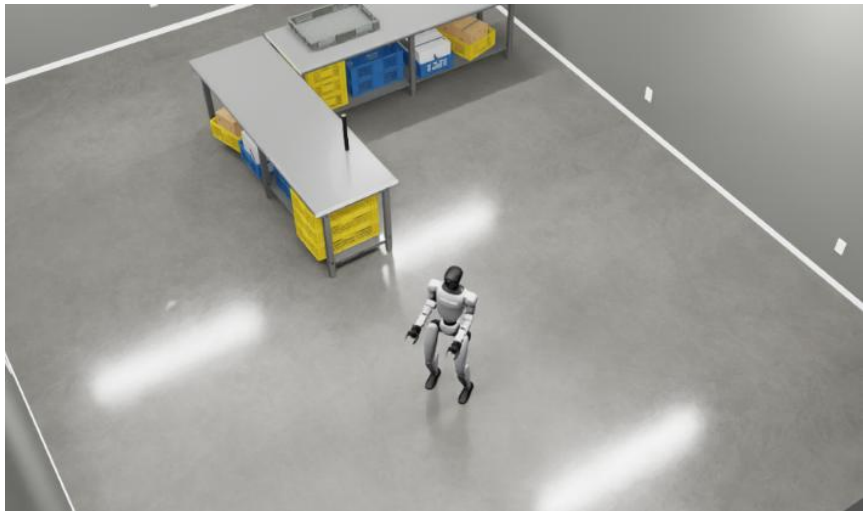


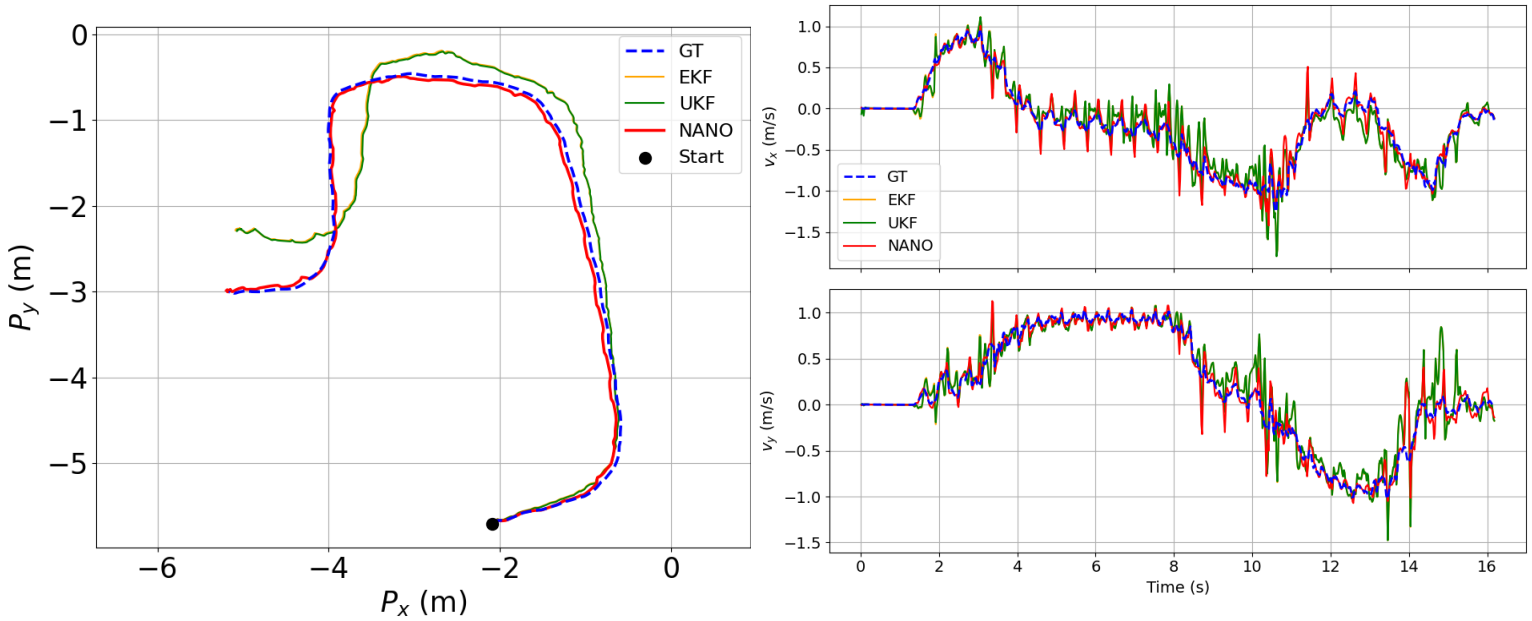
Method	RMSE [m]	Time [s]
EKF	0.373	3.905
InEKF	0.281	0.712
NANO	**0.212**	**3.351**

Task 4: Humanoid Robot Pose Estimation

Unitree G1 humanoid robot

- State $x \in \mathbb{R}^{21}$, including robot's pose, velocity, IMU bias, foots position
- Measurement $y \in \mathbb{R}^{14}$, including 2 foot-body positions, foot-body velocities and foot-world heights
- Both transition and measurement model are nonlinear





Method	ATE (pos) [m]	ATE (vel) [m]
EKF	0.386	0.346
UKF	0.387	0.345
NANO	**0.101**	**0.267**

Recent Publications

- W. Cao, T. Zhang, Z. Sun, C. Liu, S. S. Yau and S. E. Li. Nonlinear Bayesian Filtering with Natural Gradient Gaussian Approximation. *IEEE Trans on Pattern Analysis and Machine Intelligence, 2026 (online in arXiv, Oct 2024)*
- W. Cao, T. Zhang and S. E. Li. Algorithm Design and Comparative Test of Natural Gradient Gaussian Approximation Filter. IFAC *Modeling Estimation and Control Conference* (MECC), Pittsburgh, USA, 2025
- T. Zhang, W. Cao, C. Liu, F. Zhang, W. Wu and S. E. Li. NANO-SLAM: Natural Gradient Gaussian Approximation for Vehicle SLAM. IEEE *28th International Conference on Intelligent Transportation Systems (ITSC),* Gold Coast, Australia, 2025
- T. Zhang, W. Cao and S. E. Li. Natural Gradient Gaussian Approximation Filter with Positive Definiteness Guarantee. *American Control Conference* (ACC), New Orleans, USA, 2026
- T. Zhang, W. Cao, C. Liu, Y. Lyu and S. E. Li. Natural Gradient Gaussian Approximation Filter on Lie Groups for Robot State Estimation. submitted to *IEEE Trans on Control Systems Technology* (IEEE TCST), 2026
- Open-source code: https://github.com/wenhancao/NANO-idlab

Thanks for your attention!

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